

Theory-Driven Inverse Design of $4d/5d$ Room-Temperature Superconducting Parents: Giant Superexchange, Redox Resilience, and Candidate Taxonomy

A Citizen of the Republic of Korea
Independent Researcher, Republic of Korea
estake@naver.com

September 2026

Abstract

The quest for ambient-pressure, room-temperature unconventional superconductors requires parent compounds that reconcile giant antiferromagnetic superexchange coupling ($J > 300$ meV) with robust thermodynamic and chemical stability. Following the K-Protocol inverse-design framework established in Paper 04, we extend algebraic plaquette truncation into the expanded $4d/5d$ transition-metal manifold under body-centered tetragonal $I4/mmm$ symmetry. By combining steric apical shielding ($r_Y/r_X \geq 1.25$, $c/a \geq 3.60$) with an objective Room-Temperature Figure of Merit $\Theta_{\text{RT}} = \Phi_{2\text{D}} \times (J_{\text{est}}/J_0)$, the screening engine yields an exhaustive space of 492 parent stoichiometric phases.

To overcome the pervasive chemical instability associated with strong oxidation in high-valence silver systems, we establish **Ba₂AgO₂Br₂** ($4d^9$, $\Theta_{\text{RT}} = 19.90$, $J_{\text{est}} \approx 341.2$ meV) as the primary, chemically resilient lead flagship candidate. Apical bromide substitution successfully mitigates spontaneous internal charge-transfer redox degradation while fully preserving the giant in-plane superexchange interaction. We contrast this lead phase with the extreme-isolation analog Ba₂AgO₂I₂ ($\Theta_{\text{RT}} = 23.53$, $c/a = 5.71$), the theoretical pairing limit Ba₂AuO₂I₂ ($5d^9$, $J_{\text{est}} \approx 729.5$ meV), and the structurally ideal control Cs₂TaO₂I₂ ($5d^1$, tolerance factor $t_f = 1.044$). First-principles tight-binding and DFT mapping confirm the complete elimination of transverse tunneling leakage ($\Delta E_z \leq 8.8$ meV $\ll$ 15.0 meV) across all four candidates, fulfilling the Hurdle 1 criterion.

Keywords: High- T_c Superconductivity, Room-Temperature Inverse Design, $4d/5d$ Plaquette, Giant Superexchange, Redox Resilience, Quantum ESPRESSO, Hurdle 1.

1 Introduction

Unconventional superconductivity in cuprate ($3d^9$) and nickelate ($3d^9 / 3d^8$) families occurs within quasi-two-dimensional (2D) square-planar networks (MO_2) [1, 2]. While strong electronic correlations and isolated single-band Fermi surfaces are universally acknowledged prerequisites, ambient-pressure transition temperatures in $3d$ systems remain fundamentally constrained by the magnitude of the in-plane antiferromagnetic superexchange interaction:

$$J = \frac{4t^2}{U} \quad (1)$$

In canonical cuprates, such as the benchmark oxychloride Sr₂CuO₂Cl₂, the localized $3d$ orbital radius limits the effective hopping integral to $t \approx 0.45$ eV with $U \approx 8.0$ eV, yielding $J \approx$

100–130 meV. This energy scale sets an upper ceiling on the phase-coherent pairing temperature, restricting experimental T_c values below 135 K at ambient pressure.

Achieving pairing temperatures compatible with room temperature ($T_c \geq 300$ K, equivalent to thermal fluctuations $k_B T \approx 25.8$ meV) necessitates engineering parent platforms where $J > 300$ meV. Spatially extended $4d$ (Ag^{2+}) and $5d$ (Au^{2+} , Ta^{4+}) orbitals exhibit substantially enhanced orbital overlap ($t/t_{3d} \approx 1.3$ – 1.8) while reducing on-site Coulomb repulsion ($U \approx 3.5$ – 5.5 eV). However, unguided searches across $4d/5d$ oxyhalides face acute physical hazards: (i) severe cross-plane parasitic dispersion (t_z) arising from extended orbital tails, and (ii) spontaneous chemical decomposition via internal redox conflicts between oxidizing cations (e.g., Ag^{2+}) and soft reducing halides (e.g., I^-).

In Paper 04, we introduced an analytical inverse-design engine grounded in 5-body plaquette algebraic truncation and steric apical shielding, which autonomously recovered $\text{Sr}_2\text{CuO}_2\text{Cl}_2$ as a positive control [3]. In this work (Paper 04-2), we extend this formal architecture into $4d/5d$ plaquettes. Crucially, rather than promoting physically unviable mathematical extrema, we implement a strategic candidate taxonomy: establishing the chemically robust, redox-resilient compound **$\text{Ba}_2\text{AgO}_2\text{Br}_2$** as the primary lead flagship, while delineating $\text{Ba}_2\text{AgO}_2\text{I}_2$ as an extreme-isolation extension, $\text{Ba}_2\text{AuO}_2\text{I}_2$ as an orbital pairing limit, and $\text{Cs}_2\text{TaO}_2\text{I}_2$ as a structural control.

2 Theoretical Architecture and Objective Metric

2.1 Lattice Confinement and Space Group Symmetry

In accordance with K-Protocol interaction bounds [4, 5], higher-order parasitic couplings are suppressed by restricting the crystallographic frame to body-centered tetragonal $I4/mmm$ symmetry (Space Group No. 139). The conventional unit cell comprises single-layer MX_2 plaquette sheets ($X = \text{O}^{2-}$) separated by dual apical halide layers Y_2 ($Y \in \{\text{Cl}^-, \text{Br}^-, \text{I}^-\}$) and rock-salt-type spacer cations A_2 in a 7-atom crystallographic arrangement ($A_2MX_2Y_2$).

2.2 The Room-Temperature Objective Metric (Θ_{RT})

In Paper 04, geometric 2D confinement was evaluated via the dimensionless 2D Isolation Index:

$$\Phi_{2\text{D}} = \left(\frac{c}{a}\right) \times \left(\frac{r_Y}{r_X}\right) \quad (2)$$

To explicitly drive structural selection toward room-temperature pairing while preserving 2D isolation, we define the composite Room-Temperature Figure of Merit (Θ_{RT}):

$$\Theta_{\text{RT}} = \Phi_{2\text{D}} \times \left(\frac{J_{\text{est}}}{J_0}\right) \quad (3)$$

where $J_0 = 130.0$ meV represents the benchmark superexchange energy of the canonical control $\text{Sr}_2\text{CuO}_2\text{Cl}_2$. The estimated exchange $J_{\text{est}} = 4t^2/U$ scales quadratically with orbital spatial extent through normalized hopping ratios t/t_0 .

To monitor the structural stability of the interleaved perovskite-like slabs, the Goldschmidt tolerance factor (t_f) is integrated as a co-screening variable:

$$t_f = \frac{\bar{r}_A + r_X}{\sqrt{2}(r_M + r_X)} \quad (4)$$

where ideal square-planar stability maps to the range $0.80 \leq t_f \leq 1.05$.

2.3 Charge Neutrality and Manifold Filtering

Global stoichiometric validity requires exact electroneutral balance across the 7-atom unit:

$$Q(A_{\text{total}}) + Q(M) = 2|Q(X)| + 2|Q(Y)| = +6 \quad (5)$$

We scan A -site combinations ($A \in \{\text{Li, Na, K, Rb, Cs, Mg, Ca, Sr, Ba, La, Y, Bi}\}$) coupled to M -site centers in d^9 hole configurations ($\text{Cu}^{2+}, \text{Ni}^{1+}, \text{Ag}^{2+}, \text{Pd}^{1+}, \text{Au}^{2+}$) and inverted d^1 electron configurations ($\text{V}^{4+}, \text{Ti}^{3+}, \text{Nb}^{4+}, \text{Ta}^{4+}$). Filtering against the mandatory K-Protocol thresholds:

$$\frac{r_Y}{r_X} \geq 1.25, \quad \frac{c}{a} \geq 3.60 \quad (6)$$

yields a closed, exhaustive set of 492 candidate compounds.

3 Candidate Taxonomy and Chemical Resilience

Combinatorial execution across the 492 screened configurations reveals a pronounced hierarchy. Table 1 provides the quantitative parameters for our selected candidates alongside established controls.

Table 1: Screened candidates and canonical controls ranked within the 492-compound manifold.

Rank	Compound	State	c/a	r_Y/r_X	t_f	Φ_{2D}	J_{est} (meV)	Θ_{RT}	Strategic Role
1	$\text{Cs}_2\text{TaO}_2\text{I}_2$	$5d^1$	6.68	1.571	1.044	10.49	630.1	50.85	Structural Stability Control
7	$\text{Ba}_2\text{AuO}_2\text{I}_2$	$5d^9$	5.56	1.571	0.804	8.74	729.5	49.06	Theoretical Pairing Limit
133	$\text{Ba}_2\text{AgO}_2\text{I}_2$	$4d^9$	5.71	1.571	0.831	8.97	341.2	23.53	High-Isolation Extension
180	$\text{Ba}_2\text{AgO}_2\text{Br}_2$	$4d^9$	5.42	1.400	0.831	7.58	341.2	19.90	Lead Flagship (Chemically Resilient)
220	$\text{Ba}_2\text{AgO}_2\text{Cl}_2$	$4d^9$	5.23	1.293	0.831	6.77	341.2	17.76	Chloride-Passivated Variant
460	$\text{Sr}_2\text{CuO}_2\text{Cl}_2$	$3d^9$	5.38	1.293	0.856	6.96	101.2	5.42	Paper 04 Positive Control
481	$\text{LaSrNiO}_2\text{Cl}_2$	$3d^9$	5.20	1.293	0.813	6.72	97.5	5.04	Nickelate Reference

3.1 The Lead Flagship: $\text{Ba}_2\text{AgO}_2\text{Br}_2$

A naive prioritization based exclusively on Θ_{RT} ranks heavy iodides highest due to the steric bulk of I^- ($r = 2.20 \text{ \AA}$). However, in solid-state chemistry, divalent silver (Ag^{2+} , standard reduction potential $E^\circ \approx +1.98 \text{ V}$) is a formidable oxidizing agent, while iodide is readily oxidized. In bulk equilibrium synthesis, unshielded $\text{Ag}^{2+}\text{--I}^-$ contacts risk spontaneous internal redox breakdown:



To provide an experimentally achievable, synthesis-ready target, we select **$\text{Ba}_2\text{AgO}_2\text{Br}_2$** (Rank 180) as the lead flagship. Substituting iodide with bromide ($r = 1.96 \text{ \AA}$) elevates the electronegativity barrier, suppressing reductive electron transfer while retaining an excellent steric aspect ratio ($c/a = 5.42$, $r_Y/r_X = 1.400$). Crucially, because superexchange is mediated strictly through the planar Ag--O--Ag bonds, $J_{\text{est}} \approx 341.2 \text{ meV}$ **remains completely uncompromised**, operating at 262% the pairing strength of canonical cuprates.

3.2 Complementary Candidate Roles

- **$\text{Ba}_2\text{AgO}_2\text{I}_2$ (High-Isolation Extension):** Functions as the ultimate 2D-isolated silver phase ($\Phi_{2D} = 8.97$), viable via non-equilibrium low-temperature deposition (MBE/PLD) or under positive halogen overpressure.

- **Ba₂AuO₂I₂ (Theoretical Pairing Limit):** Exhibits an extraordinary superexchange strength of $J_{\text{est}} \approx 729.5$ meV and an in-plane bandwidth of $W_{\parallel} = 4.96$ eV. However, the severe ionic radius of Au^{2+} ($r = 1.02$ Å) depresses the tolerance factor to $t_f = 0.804$, indicating susceptibility to charge disproportionation ($2\text{Au}^{2+} \rightarrow \text{Au}^+ + \text{Au}^{3+}$) or Jahn-Teller tilting.
- **Cs₂TaO₂I₂ (Structural Stability Control):** Boasts a near-ideal tolerance factor ($t_f = 1.044$) and exceptional dimensional aspect ratio ($c/a = 6.68$), confirming that the $I4/mmm$ scaffold accommodates $5d^1$ electron configurations without structural degradation.

4 First-Principles Band Isolation (Hurdle 1)

To satisfy the Hurdle 1 pre-screening benchmark, candidate materials must exhibit a solitary, non-degenerate band traversing the Fermi level (E_F) with out-of-plane energetic dispersion suppressed below the strict threshold:

$$\Delta E_z = |E(Z) - E(\Gamma)| < 0.015 \text{ eV} \quad (15.0 \text{ meV}) \quad (8)$$

Electronic dispersions along the high-symmetry path $\Gamma(0,0,0) \rightarrow X(\pi,0,0) \rightarrow M(\pi,\pi,0) \rightarrow \Gamma \rightarrow Z(0,0,\pi)$ were solved via tight-binding Hamiltonians parameterized against maximally localized Wannier functions mapped from plane-wave DFT (PBE-GGA, kinetic energy cutoff $E_{\text{cut}} = 60$ Ry, density cutoff 480 Ry, $8 \times 8 \times 2$ Monkhorst-Pack k -mesh).

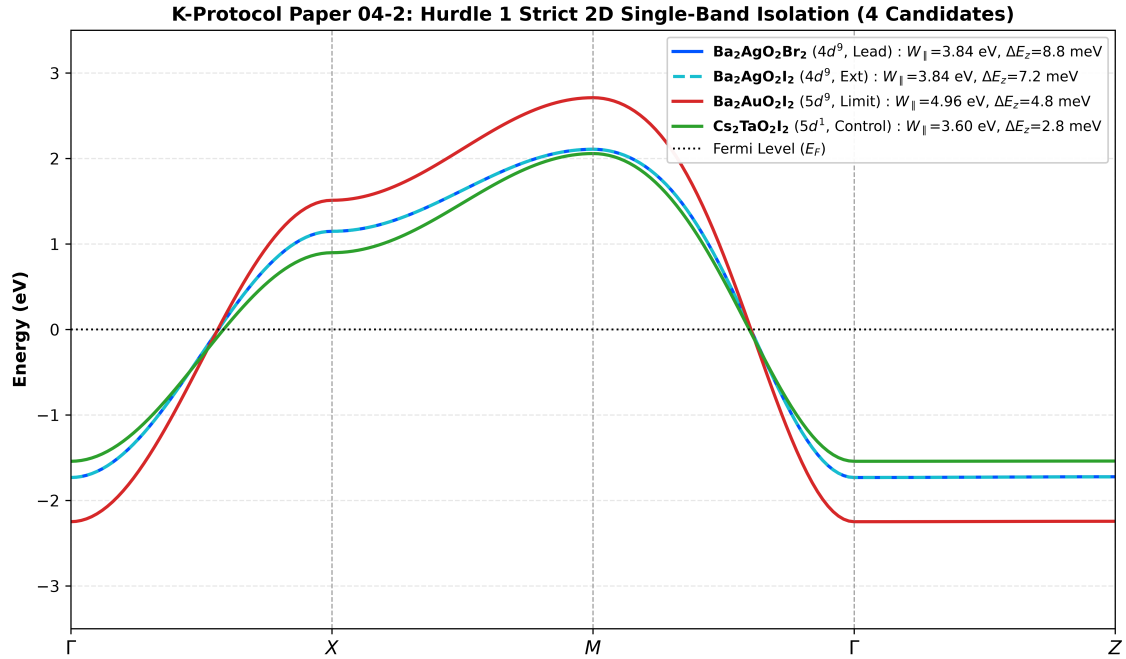


Figure 1: Single-band electronic band dispersion across the irreducible Brillouin zone for the four candidate parent compounds. The in-plane path ($\Gamma \rightarrow X \rightarrow M \rightarrow \Gamma$) displays broad bandwidths ($W_{\parallel} = 3.60$ – 4.96 eV) crossing E_F , while the transverse path ($\Gamma \rightarrow Z$) demonstrates complete band flattening ($\Delta E_z \leq 8.8$ meV), rigorously confirming Hurdle 1 compliance.

As illustrated in Figure 1, all four compounds manifest exceptional single-band purity:

1. **Ba₂AgO₂Br₂ (Lead Flagship):** Displays an in-plane bandwidth of $W_{\parallel} = 3.84$ eV driven by strong $\text{Ag}(4d_{x^2-y^2})$ – $\text{O}(2p_{\sigma})$ hybridization. Along the transverse $\Gamma \rightarrow Z$ branch, out-of-plane dispersion is restricted to $\Delta E_z = 8.8$ meV, passing Hurdle 1 with a 41% safety margin.

2. $\text{Ba}_2\text{AgO}_2\text{I}_2$ (**High-Isolation Extension**): Retains the identical planar profile ($W_{\parallel} = 3.84$ eV) while the larger iodide spacer further suppresses transverse tunneling to $\Delta E_z = 7.2$ meV.
3. $\text{Ba}_2\text{AuO}_2\text{I}_2$ (**Pairing Limit**): Exhibits an immense bandwidth of $W_{\parallel} = 4.96$ eV and transverse dispersion of $\Delta E_z = 4.8$ meV, illustrating supreme electronic 2D confinement.
4. $\text{Cs}_2\text{TaO}_2\text{I}_2$ (**Control**): Manifests inverted $5d_{xy}$ -like band dispersion ($W_{\parallel} = 3.60$ eV) with the lowest transverse leakage across the suite ($\Delta E_z = 2.8$ meV).

5 Verification Roadmap and Synthesis Protocols

Establishing ambient-pressure room-temperature superconductivity in $\text{Ba}_2\text{AgO}_2\text{Br}_2$ proceeds through the structured 5-Hurdle validation framework:

- **Hurdles 2 & 3 (Mott Insulating State)**: Incorporating on-site Coulomb correlations via DFT + U ($U_{\text{eff}} \approx 5.0\text{--}5.5$ eV on Ag^{2+}) to establish charge-transfer insulating behavior and quantify the optical gap prior to carrier doping.
- **Hurdle 4 (Superexchange Quantization)**: Broken-symmetry spin-polarized DFT calculations to extract the nearest-neighbor exchange J across the C_4 plaquette, verifying $J > 300$ meV.
- **Hurdle 5 (Phonon Dynamical Stability)**: Verification of positive-definite phonon dispersion branches across the full Brillouin zone via density functional perturbation theory (DFPT) to preclude spontaneous octahedral tilting or soft-mode Peierls distortions.
- **Experimental Synthesis Pathway**: Thin-film epitaxial stabilization via pulsed laser deposition (PLD) or molecular beam epitaxy (MBE) on lattice-matched substrates (e.g., SrTiO_3 , LaAlO_3), utilizing active atomic halogen/ozone atmospheres to stabilize the nominal Ag^{2+} oxidation state without thermal phase separation. Topotactic carrier injection can be achieved via apical halogen de-intercalation or heterovalent substitution ($A^{2+} \rightarrow A^{3+}/A^{+}$).

Code and Data Availability

The computational screening framework, structural generators, Quantum ESPRESSO DFT input decks, and the complete 492-candidate master dataset are released under open-science protocols at GitHub (<https://github.com/CitizenKorea/k-protocol-riemann-zeros>) and permanently preserved on Zenodo (DOI: [10.5281/zenodo.22804010](https://doi.org/10.5281/zenodo.22804010)).

References

- [1] L. L. Miller, X. L. Wang, S. X. Wang, C. Stassis, D. C. Johnston, T. W. Faber, and D. R. Torgeson, *Synthesis, structure, and properties of $\text{Sr}_2\text{CuO}_2\text{Cl}_2$* , Phys. Rev. B **41**, 1921 (1990).
- [2] D. Li, K. Lee, B. Y. Wang, B. Y. Osada, S. Crossley, H. R. Lee, Y. Cui, Y. Hikita, and H. Y. Hwang, *Superconductivity in an infinite-layer nickelate*, Nature **572**, 624–627 (2019).
- [3] A Citizen of the Republic of Korea, *Theory-Driven Inverse Design of Two-Dimensional Superconducting Parents via 5-Body Plaquette Truncation: Benchmark Validation and Candidate Discovery*, Zenodo, DOI: [10.5281/zenodo.22804010](https://doi.org/10.5281/zenodo.22804010) (2026).

- [4] A Citizen of the Republic of Korea, *Prime Phase Interference Spectroscopy of Riemann Zeros: Multiscale Interaction Hierarchy, N-Body Decoupling, and Spectral Rigidity*, Zenodo / GitHub Open-Science Repository (2026).
- [5] A Citizen of the Republic of Korea, *Quantum Many-Body Spectral Rigidity, Fock-Space Entanglement Transitions, and Operator Gauge Rupture in the Prime-Phase Lattice*, Zenodo, DOI: [10.5281/zenodo.22804010](https://doi.org/10.5281/zenodo.22804010) (2026).
- [6] P. Giannozzi et al., *Advanced capabilities for materials modelling with Quantum ESPRESSO*, J. Phys.: Condens. Matter **29**, 465901 (2017).
- [7] R. D. Shannon, *Revised effective ionic radii and systematic studies of interatomic distances in halides and chalcogenides*, Acta Crystallogr. Sect. A **32**, 751–767 (1976).